

MiniConGTS: A Near Ultimate Minimalist Contrastive Grid Tagging Scheme for Aspect Sentiment Triplet Extraction

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Abstract

Aspect Sentiment Triplet Extraction (ASTE) aims to co-extract the sentiment triplets in a given corpus. Existing approaches within the pretraining-finetuning paradigm tend to either meticulously craft complex tagging schemes and classification heads, or incorporate external semantic augmentation to enhance performance. In this study, we, for the first time, re-evaluate the redundancy in tagging schemes and the internal enhancement in pretrained representations. We propose a method to improve and utilize pretrained representations by integrating a minimalist tagging scheme and a novel token-level contrastive learning strategy. The proposed approach demonstrates comparable or superior performance compared to state-of-the-art techniques while featuring a more compact design and reduced computational overhead. Additionally, we are the first to formally evaluate GPT-4's performance in few-shot learning and Chain-of-Thought scenarios for this task. The results demonstrate that the pretraining-finetuning paradigm remains highly effective even in the era of large language models. The codebase is available at <https://github.com/qiaosun22/MiniConGTS>.

1 Introduction

Aspect-Based Sentiment Analysis (ABSA) aims to jointly extract opinion terms, aspect terms (targets of the corresponding opinions), and their specific sentiment polarities in a given corpus. In the milestone research by Peng et al. (2020), the compound ABSA subtasks were consolidated into the Aspect Sentiment Triplet Extraction (ASTE) task framework. For each input corpus, ASTE outputs triplets in the form (Aspect, Opinion, Polarity), where the Aspect term is the target or entity being discussed, the Opinion term is the sentiment or opinion expressed about the aspect, and Polarity indicates whether the opinion is positive,

negative, or neutral. Figure 1 illustrates the ASTE task.

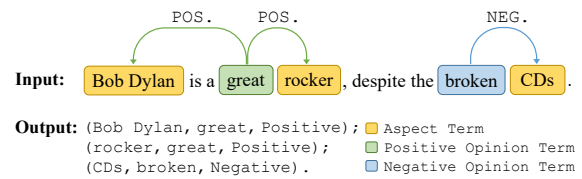


Figure 1: An illustration for ASTE, given the sentence "Bob Dylan is a great rocker, despite the broken CDs.", there are three triplets to be extracted: (Bob Dylan, great, positive), (rocker, great, positive), (CDs, broken, negative).

As an emerging fine-grained sentiment analysis initiative, ASTE offers a more detailed and nuanced understanding of sentiments in text compared to traditional methods that provide only an overall sentiment score (Peng et al., 2020). This aspect-level structured approach is inherently more challenging.

Previous approaches to ASTE have generally followed two paradigms: Pipeline methods and Joint Tagging methods (Zhang et al., 2022a). Pipeline methods decompose the ASTE task into multiple sequential subtasks, often suffering from error propagation (Xu et al., 2020). Recent progress in Machine Reading Comprehension (MRC) also contributes to this paradigm (Zhai et al., 2022; Mao et al., 2021; Zou et al., 2024; Chen et al., 2021b). Joint tagging methods adopt a unified tagging scheme to extract all triplet elements in one stage (Xu et al., 2020). The key idea is to design a **tagging scheme** (Zheng et al., 2017) that simultaneously predicts aspect terms, opinion terms, and sentiment polarities. Further developments have introduced a Grid Tagging Scheme (GTS) to represent the triplets on a unique 2D table (Wu et al., 2020a; Zhang et al., 2022b; Chen et al., 2021b, 2022; Fei et al., 2022).

Recent advances in these approaches have been focusing on the classification head design (Chen

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et al., 2022; Zhang et al., 2022b) and external semantic information enhancing (Chen et al., 2021b, 2022; Fei et al., 2022; Jiang et al., 2023; Iswari et al., 2024). However, existing research has neglected the synergistic optimization within the joint tagging scheme and the integration of contextual word representations. In this research, we propose a method to effectively improve and utilize the representation capabilities of pretrained encoders in ABSA by integrating a minimalist tagging scheme and a novel token-level contrastive learning approach.

The proposed approach demonstrates comparable or superior performance in comparison to state-of-the-art techniques, while featuring a more compact design and reduced computational overhead. Notably, even in the era of Large Language Models (LLMs), our method exhibits superior effectiveness compared to GPT 3.5 and GPT 4 in both few-shot and Chain-of-Thought (Wei et al., 2022) learning scenarios. This study provides valuable insights for the advancement of ASTE techniques within the paradigm of LLMs. Overall, our contributions are summarized as follows:

1. **Minimalist Grid Tagging Scheme:** We propose a novel minimalist joint tagging scheme that uses the fewest label classes to date.
2. **Token-level Contrastive Learning Strategy:** We introduce a token-level contrastive learning framework that enhances the contextual embeddings produced by the pretrained model. This framework is seamlessly geared towards our minimalist Grid Tagging Scheme (GTS) to effectively address the ASTE task.
3. **Comprehensive Evaluation:** We conduct extensive experiments and evaluations on multiple benchmark datasets, demonstrating the effectiveness and superiority of our proposed methods over existing approaches. Notably, we are the first to reveal GPT-4’s performance on this task, showcasing our method’s superior efficiency and effectiveness in the era of large language models.

2 Literature Review

2.1 ASTE Paradigms

Peng et al. (2020) proposed a pipeline method that divides ASTE tasks into two stages: initially extracting (Aspect, Opinion) pairs and subsequently predicting sentiment polarity. However,

pipeline methods typically suffer from error propagation issues (Xu et al., 2020). Recent pipeline methods treat ASTE as a Machine Reading Comprehension problem, and develops seq2seq methods such as machine reading comprehension (Zhai et al., 2022; Mao et al., 2021; Zou et al., 2024; Chen et al., 2021b). Joint Tagging strategies are remarked by certain Unified Tagging Scheme designs, where elements of a triplet can be extracted simultaneously. ET (Xu et al., 2020) introduced a position-aware tagging scheme with a conditional random field module, effectively addressing span overlapping issues. Recent joint paradigm methods have refined the ASTE task with the development of proficient Grid Tagging Schemes (GTS).

2.2 Grid Tagging Scheme

Wu et al. (2020a) pioneered the adoption of a grid tagging scheme (GTS) for ASTE, yielding substantial performance gains. Subsequent research refined and enhanced GTS. BDTF (Zhang et al., 2022b) designed a boundary-driven tagging scheme, effectively reducing boundary prediction errors. Alternative research augmented GTS by integrating external semantic information as structured knowledge into their models. S³E² (Chen et al., 2021b) retained the GTS tagging scheme while introducing novel semantic and syntactic enhancement modules between word embedding outputs and the tagging scheme. EMGCN (Chen et al., 2022) incorporated external knowledge from four areas—Part-of-Speech Combination, Syntactic Dependency Type, Tree-based Distance, and Relative Position Distance—through an exogenous hard-encoding strategy. SyMux (Fei et al., 2022) contributed a unified tagging scheme capable of handling all ABSA subtasks by integrating insights from GCN, syntax encoders, and representation multiplexing.

2.3 Contrastive Learning

While contrastive learning has gained popularity in diverse NLP domains (Wu et al., 2020b; Giorgi et al., 2021; Gao et al., 2021; Zhang et al., 2021), its application to ASTE remains relatively unexplored. Ye et al. (2021) adopts contrastive learning into triplet extraction in a generative fashion. Wang et al. (2022) takes contrastive learning as a data augmentation approach. Yang et al. (2023) proposed an enhancement approach in pairing with two separate encoders.

3 Method

3.1 Overall Framework

An overall description of the training process can be found in Figure 2. Basically our design can be break down into the **Minimalist Grid Tagging Scheme (GTS)** and the **Token-level Contrastive Learning Strategy**.

Tokenize the input sequence S using the Tokenizer Tk and pass the tokenized sequence through the Pretrained Language Model PLM (such as BERT) to obtain contextualized representations h :

$$h = \text{PLM}(\text{Tk}(S)). \quad (1)$$

Then, the inference phase involves with forming the Minimalist GTS and predicting the corresponding class for each cell. Once the GTS is predicted, it can be decoded by the GTS decoder into the triplets in natural language form. The training phase additionally introduces a novel contrastive learning strategy, where similar and dissimilar pairs of contextual representations are distinguished. The contrastive loss is then weighted and summed with the tagging loss, which is the classification loss between the predicted and ground truth tagging schemes.

Our research benefits from the following two closely intertwined aspects: 1) The use of the Minimalist GTS simplifies the learning process by reducing the number of labels, facilitating faster convergence and seamlessly gearing the contrastive learning. 2) The token-level contrastive learning enhances the model’s ability to distinguish between related and unrelated elements within the input sequence, thereby improving the overall accuracy of the tagging system. For a more detailed description for our algorithm pipeline, see the pseudo code in Appendix A.1.

3.2 Minimalist Grid Tagging Scheme

3.2.1 Tagging Scheme Design

As defined by Section 3.1, once an input sentence is encoded into a sequence of contextual representations $h = \{h_1, h_2, \dots, h_{|h|}\}$, we form a $|h| \times |h|$ matrix, that is, our tagging scheme $\text{tag}_{|h| \times |h|}$. As shown in Figure 3, on the rows we mark Aspect tokens by yellow and the columns we mark Opinion tokens by green (positive) and blue (negative). Then the each intersection of these marked rows and columns can uniquely represent an identical sentiment triplet. Thus, each such triplet can be noted by a 2-D **area** (submatirx) in the matrix,

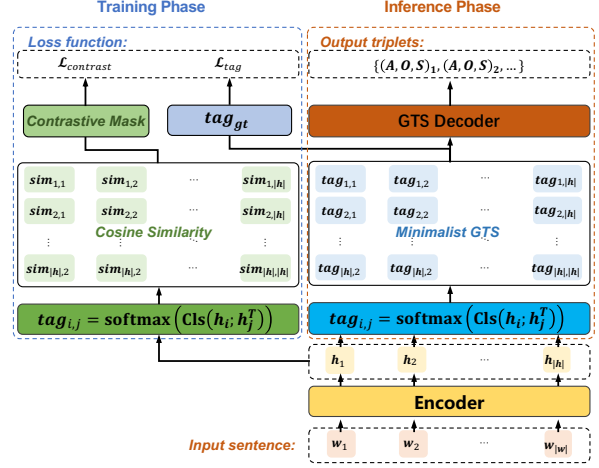


Figure 2: An overview of the proposed method, where the “Encoder” denotes for the sequential combination of a Tokenizer and a Pretrained Language Model (PLM).

where Sentiment Polarity is indicated with POS. (positive), NEU. (neutral), or NEG. (negative) in the top-left corner cell of the area, while CTD. indicates the continuation of the pairing relationship within the same region. MSK. (mask) on the diagonal represents masked cells that are not involved in the computation. In Figure 3, an example sentence is tokenized and tagged.

By defining our grid tagging scheme, we frame the triplet extraction problem as a 5-class classification task, using the fewest number of labels known to date. In Appendices A.3 - A.5, we provide rigorous proof and heuristic insights to justify our design and ensure its rationality.

3.2.2 Tagging Loss

We adopt a tagging loss to guide neural network learning.

We concatenate the representation with its transposed form to construct a matrix. Then, we apply the classification head Cls to the embeddings, followed by the softmax function, to obtain the predicted classification probabilities for each cell:

$$\hat{\text{tag}}_{i,j} = \text{softmax}(\text{Cls}(h_i; h_j^T)), i, j = 1, \dots, |h|. \quad (2)$$

The focal loss (Lin et al., 2017) is employed to mitigate class imbalance by placing greater emphasis on examples that are difficult to classify correctly. This is achieved by down-weighting the loss for well-classified instances and focusing more on misclassified instances. The formula for focal

	Bob	Dylan	is	great	rock	#er	but	brok	#en	CD	#s
Bob	MSK.			POS.							
Dylan		MSK.		CTD.							
is			MSK.								
great				MSK.							
rock				POS.	MSK.						
#er				CTD.		MSK.					
but							MSK.				
brok								MSK.			
#en									MSK.		
CD										NEG.	CTD.
#s										CTD.	CTD.

Figure 3: The grid tagging scheme employs the fewest classes of labels while completely handle all the triplet cases without conflict, overlap or omission. Each **area** circled in red dashed lines corresponds to a triplets. For example, intersection area between columns of "broken" and rows of "CDs" is marked as negative, with NEG. on its top-left cell and CTD. for others. It is worth mentioning that the blank cells in the matrix are labeled as an additional class but are omitted for visual simplicity.

loss $\mathcal{L}$ is as follows:

$$\mathcal{L}_{\text{tag}} = -\frac{1}{|h|^2} \sum_{i,j=1}^{|h|} \alpha_{\text{tag}_{i,j}} (1 - \text{tag}_t)^\gamma \log(\text{tag}_t), \quad (3)$$

where α is a weighting factor for balancing the importance of tags, γ is a focusing parameter that increases the weight of hard-to-predict tags, and $\text{tag}_{i,j}$ and tag_t represent the ground truth label and the predicted probability for the true label at position (i, j) , respectively:

$$\text{tag}_t = \hat{\text{tag}}_{i,j;\text{tag}_{i,j}} \quad (4)$$

3.3 Contrastive Learning Strategy

3.3.1 Contrastive Learning Label Matrix

Contrastive learning is an unsupervised learning method that aims to learn effective feature embeddings by pulling together similar pairs of samples and pushing apart dissimilar pairs. In our design, we construct a label matrix where each cell is annotated by either PULL or PUSH, which means making the representations closer among tokens within the same class and farther between those of different classes. See an illustration of this strategy in Figure 4.

	Bob	Dylan	is	great	rock	#er	but	brok	#en	CD	#s
Bob	MSK.	PULL	PUSH	PUSH	PULL	PULL	PUSH	PUSH	PUSH	PULL	PULL
Dylan	MSK.	MSK.	PUSH	PUSH	PULL	PULL	PUSH	PUSH	PUSH	PULL	PULL
is	MSK.	MSK.	MSK.	PUSH	PUSH	PUSH	PULL	PUSH	PUSH	PUSH	PUSH
great	MSK.	MSK.	MSK.	MSK.	PUSH	PUSH	PUSH	PUSH	PUSH	PUSH	PUSH
rock	MSK.	MSK.	MSK.	MSK.	MSK.	PULL	PUSH	PUSH	PUSH	PULL	PULL
#er	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	PUSH	PUSH	PUSH	PULL	PULL
but	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	PUSH	PUSH	PUSH	PUSH
brok	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	PULL	PUSH	PUSH
#en	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	PUSH	PUSH
CD	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	PULL
#s	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.	MSK.

Figure 4: An illustration for the ‘‘Contrastive Mask’’. Each token is paired with every other token, where PULL denotes positive sample pairs, indicating that the tokens belong to the same category and should be pulled closer together, while PUSH denotes negative sample pairs, indicating that the tokens belong to different categories and should be pushed apart. The lower triangular part of the matrix, marked by MSK. are masked cells that are not involved in the computation. For example, ‘‘Bob’’ and ‘‘Dylan’’ are marked as a positive sample pair with PULL, indicating similarity, while ‘‘Bob’’ and ‘‘is’’ are marked as a negative sample pair with PUSH, indicating dissimilarity.

3.3.2 Objective Function

The commonly used InfoNCE (Information Noise-Contrastive Estimation) loss function (van den Oord et al., 2019) is employed:

$$\mathcal{L}_{\text{contrast}} = -\sum_{i=1}^N \log \frac{\exp(\text{sim}(\mathbf{h}_i, \mathbf{h}_i^+))}{\exp(\text{sim}(\mathbf{h}_i, \mathbf{h}_i^+)) + \sum_{j=1}^M \exp(\text{sim}(\mathbf{h}_i, \mathbf{h}_i^-))}, \quad (5)$$

where $\mathbf{h}_i^+ / \mathbf{h}_j^-$ represents the positive / negative sample embedding with the anchor respectively. $\text{sim}(\cdot, \cdot)$ denotes the similarity function, which is calculated by the cosine similarity:

$$\text{sim}(\mathbf{h}_i, \mathbf{h}_i) = \frac{\mathbf{h}_i \cdot \mathbf{h}_i}{\|\mathbf{h}_i\| \|\mathbf{h}_i\|} \quad (6)$$

3.4 Overall Loss Function

The overall loss $\mathcal{L}$ can be formulated as a weighted sum of two individual loss functions: the tagging loss $\mathcal{L}_{\text{tag}}$ and the contrastive loss $\mathcal{L}_{\text{contrast}}$:

$$\mathcal{L} = \mathcal{L}_{\text{tag}} + \beta \mathcal{L}_{\text{contrast}}, \quad (7)$$

where β is a coefficient for balancing the different parts of the loss. This combination allows for balancing the influence of each component in the training process.

4 Experiments

4.1 Implementation Details

All experiments were performed on a single RTX 2080 Ti. The best model weight on the development set is saved and then evaluated on the test set. For the PLM encoder, the pretrained weights `bert_base_uncased` and `roberta_base` are downloaded from (Wolf et al., 2020). GPT 3.5-Turbo and GPT 4 are implemented using OpenAI API (OpenAI, 2024). The learning rate is 1×10^{-5} for the PLM encoder, and 1×10^{-3} for the classification head.

4.2 Datasets

We evaluate our method on two canonical ASTE datasets derived from the SemEval Challenges (Pontiki et al., 2014, 2015, 2016). These datasets serve as benchmarks in most aspects-based sentiment analysis (ABSA) research. The first dataset, denoted as $\mathcal{D}_1$, is the Aspect-oriented Fine-grained Opinion Extraction (AFOE) dataset introduced by (Wu et al., 2020a). The second dataset, denoted as $\mathcal{D}_2$, is a refined version by (Xu et al., 2020), building upon the work of (Peng et al., 2020). More details are provided in Appendix A.2.

4.3 Baselines

We evaluate our method against various techniques including pipeline, sequence-labeling, seq2seq, table-filling and LLM-based approaches. Detailed descriptions for each method can be found in the Appendix A.6.

4.4 ASTE Performance

4.4.1 Comparison to Existing Methods

We evaluate ASTE performance using the widely accepted (Precision, Recall, F1) metrics. Results of the dataset $\mathcal{D}_2$ are shown in Table 1, while the results of $\mathcal{D}_1$ are presented in Appendix Table 8. The best results are highlighted in bold, and the second-best results are underscored. Our proposed method consistently achieves state-of-the-art performance or ranks second in most evaluated cases.

Notably, on dataset $\mathcal{D}_1$, the proposed method achieves a substantial 3.08% improvement in F1 score on the 14Lap subset. This improvement is

particularly significant given that the highest score on this subset is the lowest among all datasets, showcasing our method’s effectiveness in handling challenging instances. Moreover, on the 14Res subset, our F1 score exceeds 76.00, which, to the best of our knowledge, is the highest reported performance. For dataset $\mathcal{D}_2$, our method outperforms all state-of-the-art approaches by over 1 percentage point on the 14Res, 14Lap, and 16Res subsets. Only on the 16Res subset does the BDTF method (Zhang et al., 2022b) achieve a slightly better performance.

4.4.2 Comparison to GPT

Our proposed method is based on the Pretrain-Finetuning paradigm, which is increasingly challenged by large language models (LLMs) (Kojima et al., 2022; Wei et al., 2021). It is concerned about how the advancing capabilities of LLMs might impact the ASTE task.

When compared to advanced LLMs, the performance and computational efficiency of our method stand out. As shown in Tables 8, 1, and 9, even the state-of-the-art LLM, GPT-4, with its staggering number of parameters, does not achieve satisfactory results for ASTE, even with few-shot learning and Chain-of-Thought (CoT) (Wei et al., 2022) enhancement. Additionally, using LLMs introduces significant computational overhead. For more information on experiment setting see Appendix A.9. For detailed results see Table 1, 12, Appendix A.8 and Appendix A.10. Note that, fine-tuning LLMs may offer some improvements, but it also risks catastrophic forgetting (Shi et al., 2024) and is left for future work.

To our knowledge, this is the first formal study to evaluate GPT-4’s performance on these ASTE datasets, providing valuable insights for future research.

4.5 Performance on Other ABSA Tasks

Our method can also effectively handle other ABSA subtasks, including Aspect Extraction (AE), Opinion Extraction (OE), and Aspect Opinion Pair Extraction (AOPE). AE aims to extract all the (Aspect) terms, OE aims to extract all the Opinion terms, and AOPE aims to extract all the (Aspect, Opinion) pairs from raw text. The results for these tasks are presented in Appendix A.7, where our method consistently achieves best F1-scores across nearly all tasks.

Methods	14Res			14Lap			15Res			16Res		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Pipeline												
Two-stage ^b (Peng et al., 2020)	43.24	63.66	51.46	37.38	50.38	42.87	48.07	57.51	52.32	46.96	64.24	54.21
Li-unified-R+PD [‡] (Peng et al., 2020)	40.56	44.28	42.34	41.04	67.35	51.00	44.72	51.39	47.82	37.33	54.51	44.31
Sequence-tagging												
Span-BART (Yan et al., 2021)	65.52	64.99	65.25	61.41	56.19	58.69	59.14	59.38	59.26	66.60	68.68	67.62
JET (Xu et al., 2020)	70.56	55.94	62.40	55.39	47.33	51.04	64.45	51.96	57.53	70.42	58.37	63.83
Seq2seq												
Dual-MRC (Mao et al., 2021)	71.55	69.14	70.32	57.39	53.88	55.58	63.78	51.87	57.21	68.60	66.24	67.40
BMRC [†] (Chen et al., 2021a)	72.17	65.43	68.64	65.91	52.15	58.18	62.48	55.55	58.79	69.87	65.68	67.35
COM-MRC (Zhai et al., 2022)	75.46	68.91	72.01	62.35	58.16	60.17	68.35	61.24	64.53	71.55	71.59	71.57
Triple-MRC (Zou et al., 2024)	-	-	72.45	-	-	60.72	-	-	62.86	-	-	68.65
Table-filling												
GTS (Wu et al., 2020a)	67.76	67.29	67.50	57.82	51.32	54.36	62.59	57.94	60.15	66.08	66.91	67.93
Double-encoder (Jing et al., 2021)	67.95	71.23	69.55	62.12	<u>56.38</u>	59.11	58.55	60.00	59.27	70.65	70.23	70.44
EMC-GCN (Chen et al., 2022)	71.21	72.39	71.78	61.70	56.26	58.81	61.54	62.47	61.93	65.62	71.30	68.33
BDTF (Zhang et al., 2022b)	75.53	<u>73.24</u>	<u>74.35</u>	68.94	55.97	61.74	68.76	<u>63.71</u>	66.12	71.44	<u>73.13</u>	72.27
STAGE-1D (Liang et al., 2023)	79.54	68.47	73.58	<u>71.48</u>	53.97	61.49	72.05	58.23	64.37	78.38	69.10	<u>73.45</u>
STAGE-2D (Liang et al., 2023)	78.51	69.3	73.61	70.56	55.16	61.88	<u>72.33</u>	58.93	64.94	<u>77.67</u>	68.44	72.75
STAGE-3D (Liang et al., 2023)	<u>78.58</u>	69.58	73.76	71.98	53.86	61.58	73.63	57.9	64.79	76.67	70.12	73.24
DGCNAP (Li et al., 2023)	72.90	68.69	70.72	62.02	53.79	57.57	62.23	60.21	61.19	69.75	69.44	69.58
LLM-based												
GPT-3.5-turbo zero-shot	44.88	55.13	49.48	30.04	41.04	34.69	36.02	53.40	43.02	39.92	57.78	47.22
GPT-3.5-turbo few-shot	51.51	65.19	57.55	39.79	50.09	44.35	43.34	63.09	51.39	51.12	71.01	59.45
GPT-3.5-turbo CoT	48.47	59.05	53.24	30.48	40.30	34.71	39.51	56.70	46.57	44.03	63.81	52.10
GPT-3.5-turbo CoT+few-shot	49.41	59.15	53.85	33.78	42.33	37.57	39.02	56.08	46.02	46.49	66.93	54.86
GPT-4o zero-shot	32.99	38.13	35.37	17.81	22.55	19.90	27.85	37.73	32.05	32.17	43.00	36.80
GPT-4o few-shot	54.11	66.20	59.55	38.23	48.61	42.80	45.57	60.41	51.95	52.90	71.01	60.63
GPT-4o CoT	41.21	53.32	46.49	26.98	37.71	31.46	33.07	50.93	40.10	39.14	58.17	46.79
GPT-4o CoT+few-shot	46.81	59.86	52.54	29.71	40.85	34.40	35.08	53.81	42.47	41.53	61.09	49.45
Ours												
MiniConGTS	76.1	75.08	75.59	66.82	60.68	63.61	66.50	63.86	<u>65.15</u>	75.52	74.14	74.83

Table 1: Experimental results on $\mathcal{D}_2$ (Xu et al., 2020). The best results are highlighted in bold, while the second best results are underscored. The results with [†] are retrieved from (Yu Bai Jian et al., 2021). The results with [‡] are retrieved from (Xu et al., 2020). The results with [‡] are retrieved from (Peng et al., 2020). The results with [‡] are retrieved from (Mao et al., 2021).

Models	$\mathcal{D}_1$				$\mathcal{D}_2$			
	14Res	14Lap	15Res	16Res	14Res	14Lap	15Res	16Res
MiniConGTS	76.00	64.07	65.43	71.80	75.59	63.61	65.15	74.83
w/o. RoBERTa	74.12	63.18	62.95	69.41	72.66	62.15	63.25	70.71
ΔF_1	-1.88	-0.89	-2.48	-2.39	-2.93	-1.46	-1.90	-4.12
w/o. contr	72.61	61.94	58.14	68.16	71.72	61.49	58.11	68.03
ΔF_1	-3.39	-2.13	-7.29	-3.64	-3.87	-2.12	-7.04	-6.80
w/o. tag	67.78	54.98	60.75	62.62	65.83	54.98	58.73	67.63
ΔF_1	-8.22	-9.09	-4.68	-9.18	-9.76	-8.63	-6.42	-7.20

Table 2: Ablation study on F1, where “w/o. RoBERTa” denotes “Replace RoBERTa with bert-base-uncased”, “w/o. contr” denotes without the contrastive learning mechanism, and “w/o. tag” denotes “replace our tagging scheme with a baseline”.

Method	Num Tags	Features Enhancing
GTS (Wu et al., 2020a)	6	None
Double-encoder (Jing et al., 2021)	9	None
EMC-GCN (Chen et al., 2022)	10	4 Groups
BDTF (Zhang et al., 2022b)	$2 \times 2 \times 3$	None
STAGE (Liang et al., 2023)	$2 \times 2 \times 4$	None
DGCNAP (Li et al., 2023)	6	POS-tagging
Ours	5	None

Table 3: Tagging Scheme Comparison.

5 Analysis

5.1 Ablation Study

In this section, we conduct a series of ablation experiments to demonstrate the superiority of our method and eliminate potential confounding factors. Experiments were conducted on the $\mathcal{D}_1$ and $\mathcal{D}_2$ datasets, using F1 scores as the comparison metric.

Encoder. We replaced the RoBERTa encoder with BERT, resulting in a slight decrease in F1 scores on both datasets, although our method still outperformed most other approaches.

Contrastive Learning. We deactivated the contrastive mechanism in our method (denoted as “w/o. contr”) by setting the coefficient of the contrastive loss to 0. The results in Table 2 illustrate a significant F1-score decrease of $2.12 \sim 7.29\%$ in both datasets.

Tagging Scheme. We substituted our proposed scheme with the conventional GTS tagging scheme

(Wu et al., 2020a), resulting in a substantial performance decline (Table 2) of 4.68 ~ 9.18%. This indicates that the contrastive learning methods, within our framework, is of strong reliance on an appropriate tagging scheme. This reinforces the effectiveness of our compact yet impactful tagging scheme.

5.2 Effect of Contrastive Learning

In Figure 5, an example is shown illustrating how contrastive learning improves representation. The right upper-row subplots show the representation outputs with contrastive learning, while the lower row subplots display that without contrastive learning. Principal Component Analysis (PCA) (Maćkiewicz and Ratajczak, 1993) is used to reduce the vector dimensions to three for visualization purposes. The distributions indicate that contrastive learning significantly enhances the representations, with similar classes of hidden word representations becoming more tightly clustered and dissimilar classes more distinct.

5.3 Efficiency Comparison

We compared the computational efficiency of Mini-ConGTS with other approaches, including baseline ASTE methods and LLMs, on an ASTE task. Evaluation metrics such as memory usage, number of parameters, epoch/inference time, and F1 scores are recorded in Table 4. Our approach not only requires less memory usage for higher performance compared to traditional ASTE methods but also offers much faster runtime even using a relatively lower-cost GPU.

Table 3 provides another comparative analysis of tagging schemes. Our method has a compact design with the fewest classes of labels. What’s more, compared with other SOTA baselines, our method does not rely on any additional linguistic information enhancement.

5.4 Case Study

A case analysis is presented in Table 12, where the proposed method demonstrates solid performance. Despite minor faults in missing the full terms, it exhibits a profound understanding of the case.

It is quite interesting to investigate the GPT’s error cases. The findings reveal that the performance of the GPT model is mixed - while it is able to identify more aspect-opinion (A-O) pairs than the ground truth annotations in some cases, this comes at the cost of a decreased precision. This suggests

that the GPT model may be “over-interpreting” the input, making inferences that go beyond what is strictly supported by the text. Furthermore, the GPT model appears to be overly sensitive to the presence of adverbs (such as “very”, “a bit”, etc.) in the input. This sensitivity manifests in the model frequently adding or removing adverbs when extracting the Opinion components, which further contributes to a decrease in the overall accuracy of the ABSA task.

These findings highlight the importance of developing ABSA models that can strike the right balance between extracting all relevant aspect-opinion pairs, while still maintaining a high degree of precision. The effective use of encoding appears to be a promising direction for achieving this balance and advancing the state-of-the-art in Aspect-Based Sentiment Analysis.

6 Conclusion

In this work, we have introduced an elegant and efficient framework for ASTE, achieving SOTA performance. Our approach is built upon two effective components: a new tagging scheme and a novel token-level contrastive learning implementation. The ablation study demonstrates the synergy between these components, reducing the need for complex model designs and external information enhancements.

7 Limitations

Our method is based on a 2D-matrix tagging scheme, where the time complexity for decoding, given the input corpus length N , is $O(N^2)$. This may be unacceptable when N is too large. Additionally, although we have demonstrated our method on commonly used classic English datasets, it should be tested on more natural corpora and for its cross-language capability.

8 Ethics & Potential Risks Statement

In our experiments, we used widely accepted datasets focused on e-commerce reviews, which have a lower risk of offensive content. We scrutinized the data for biases against gender, race, and marginalized groups. Despite these precautions, our model might still generate potentially offensive sentiment assessments if used inappropriately, such as evaluating ethical or moral statements. We reserve the right to limit or modify the use of our technology to prevent misuse.

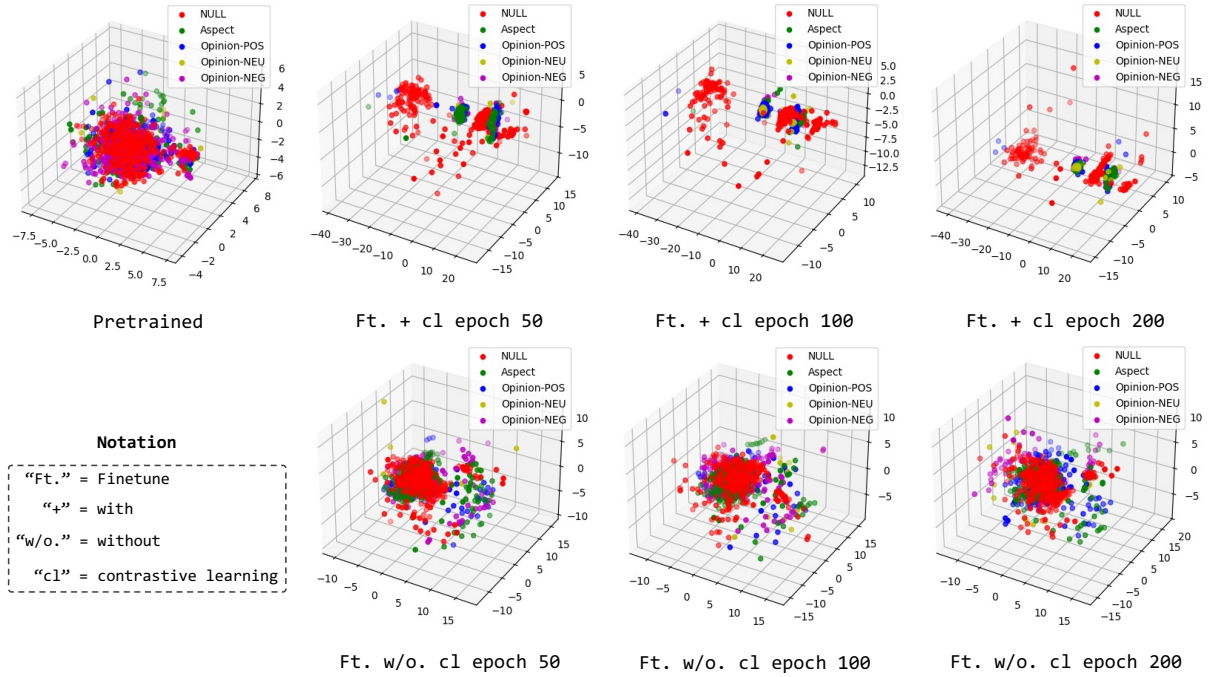


Figure 5: A plot of the hidden word representation based on the $\mathcal{D}_1$ 14Res dataset, where the dimension is reduced to 3. “Pretrained” refers to the representation output by official released model. We finetune the pretrained model with and without contrastive learning strategy respectively.

Model	Memory	Num Params	Epoch Time [‡]	Inf Time	F1(%)	Device
Span-ASTE (Xu et al., 2021)	3.173 GB [‡]	-	108s	-	71.62	Tesla v100 32GB
BDTF (Zhang et al., 2022b)	8.103 GB [‡]	>0.18B [‡]	135s	-	74.73	Tesla v100 32GB
GPT 3.5-Turbo (OpenAI, 2024)	>80 GB [‡]	175B [†]	-	0.83s	49.48	OpenAI API
GPT 4 (OpenAI, 2024)	>80 GB [‡]	1760B [‡]	-	1.56s	35.37	OpenAI API
Ours	7.11GB	0.12B	10s	0.01s	76.00	2080 Ti 11GB

Table 4: An efficiency comparison, where [†] is evaluated by (Gao, 2021) and later confirmed by OpenAI (Wikipedia, 2024), [‡] is estimated by (Schreiner, 2023), [‡] is cited from (Zhang et al., 2022b), and [‡] is reported by (Wikipedia, 2024). [‡] Epoch Time refers to the training time per epoch on the training set.

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A Appendix

A.1 Pseudo-code for the training process.

See Algorithm 1.

A.2 Descriptive Statistics of The Datasets

See Table 5.

Algorithm 1 .

Modules:

Input:

Raw sentences: $\mathcal{S}_{|\mathcal{S}|}$;

Ground truth triplets: $\mathcal{T}_{|\mathcal{T}|}^{gt}$, where

$\mathcal{T}_k = (A_k, O_k, S_k), k \in \{1, 2, \dots, |\mathcal{T}|\}$;

classes of contrasted labels: $\mathcal{C}$.

Output:

Predicted Triplets: $\mathcal{T}_{|\mathcal{T}|}^{pred}$;

Metric: *Precision, Recall, F1*.

Algorithm:

Repeat for N epochs:

1: Hidden word representation:

$\mathcal{H}_{|\mathcal{H}|} = \text{PLMsEncoder}(\mathcal{S}_{|\mathcal{S}|})$;

2: Tensor Operations:

$\mathcal{H}_{|\mathcal{H}| \times |\mathcal{H}|} = \text{expand}(\mathcal{H}_{|\mathcal{H}|})$,

$\mathcal{H}_{|\mathcal{H}| \times |\mathcal{H}|}^T = \mathcal{H}_{|\mathcal{H}| \times |\mathcal{H}|} \cdot \text{transpose}()$;

3: Similarity matrix:

$\text{Sim}_{|\mathcal{H}| \times |\mathcal{H}|} =$

$-(\mathcal{H}_{|\mathcal{H}| \times |\mathcal{H}|} - \mathcal{H}_{|\mathcal{H}| \times |\mathcal{H}|}^T) \circ (\mathcal{H}_{|\mathcal{H}| \times |\mathcal{H}|} - \mathcal{H}_{|\mathcal{H}| \times |\mathcal{H}|}^T)$
where

$$\text{Sim}_{i,j} = -\|\mathcal{H}_i - \mathcal{H}_j\|^2$$

and $\circ$ denotes the Hadamard product.

4: Contrastive Mask matrix: $\mathbf{M}_{|\mathcal{H}| \times |\mathcal{H}|}$, where $\mathbf{M}_{i,j} = 1$ if $\mathcal{H}_i, \mathcal{H}_j \in \mathcal{C}_p, p \in 1, 2, 3$ else -1 ;

5: Contrastive loss:

$$\mathcal{L}_{\text{contrastive}} = \sum_{i=1}^{|\mathcal{H}|} \sum_{j=1}^{|\mathcal{H}|} (\text{Sim}_{|\mathcal{H}| \times |\mathcal{H}|} \circ \mathbf{M}_{|\mathcal{H}| \times |\mathcal{H}|})_{i,j};$$

6: Predicted tagging matrix:

$\text{Tag}_{|\mathcal{H}| \times |\mathcal{H}|}^{pred} = \text{ClsHead}(\mathcal{H}_{|\mathcal{H}| \times |\mathcal{H}|}, \mathcal{H}_{|\mathcal{H}| \times |\mathcal{H}|}^T)$;

7: Focal loss:

$$\mathcal{L}_{\text{focal}} = \text{FocalLoss}(\text{Tag}_{|\mathcal{H}| \times |\mathcal{H}|}^{pred}, \text{Tag}_{|\mathcal{H}| \times |\mathcal{H}|}^{gt});$$

8: Weighted Loss: $\mathcal{L} = \mathcal{L}_{\text{focal}} + \alpha \mathcal{L}_{\text{contrastive}}$.

9: Backward propagation.

Predicted triplets:

$\mathcal{T}_{|\mathcal{T}|}^{pred} = \text{TaggingDecoder}(\text{Tag}_{|\mathcal{H}| \times |\mathcal{H}|}^{pred})$

Metric:

$\text{Precision, Recall, F1} = \text{Metric}(\mathcal{T}_{|\mathcal{T}|}^{pred}, \mathcal{T}_{|\mathcal{T}|}^{gt})$

A.3 Rethinking the GTS

Rethinking the 2D tagging scheme:

Lemma 1. Specific to the ASTE task, when we take it as a 2D-labeling problem, we are to 1) find a set of tagging strategies to establish a 1-1 map between each triplet and its corresponding tagging matrix. See the proof in **Appendix Proof 1**.

Lemma 2. In a 2D-tagging for ASTE, at least three basic goals must be met: 1) correctly identifying the (Aspect, Opinion) pairs, 2) correctly classifying the sentiment polarity of the pair based on the context, and 3) avoiding boundary errors, such as *overlapping*^{*}, *confusion*[†], and *conflict*[‡]. See the

^{*}It occurs when one single word belongs to multiple classes in different triplets.

[†]It occurs when there is a lack of location restrictions so that multiple neighbored candidates can not be uniquely distinguished.

[‡]It occurs when one single word is composed of multiple tokens, and the predict gives predictions that are not aligned

Datasets		#S	#A	#O	#S1	#S2	#S3	#T	
14Res	$\mathcal{D}_1$	Train	1259	1008	849	1456	164	446	2066
		Dev	315	358	321	352	44	93	489
		Test	493	591	433	651	59	141	851
	$\mathcal{D}_2$	Train	1266	986	844	1692	166	480	2338
		Dev	310	396	307	404	54	119	577
		Test	492	579	437	773	66	155	994
14Lap	$\mathcal{D}_1$	Train	899	731	693	691	107	466	1264
		Dev	225	303	237	173	42	118	333
		Test	332	411	330	305	62	101	468
	$\mathcal{D}_2$	Train	906	733	695	817	126	517	1460
		Dev	219	268	237	169	36	141	346
		Test	328	400	329	364	63	116	543
15Res	$\mathcal{D}_1$	Train	603	585	485	668	24	179	871
		Dev	151	182	161	156	8	41	205
		Test	325	353	307	293	19	124	436
	$\mathcal{D}_2$	Train	605	582	462	783	25	205	1013
		Dev	148	191	183	185	11	53	249
		Test	322	347	310	317	25	143	485
16Res	$\mathcal{D}_1$	Train	863	775	602	890	43	280	1213
		Dev	216	270	237	224	8	66	298
		Test	328	342	282	360	25	72	457
	$\mathcal{D}_2$	Train	857	759	623	1015	50	329	1394
		Dev	210	251	221	252	11	76	339
		Test	326	338	282	407	29	78	514

Table 5: Statistic information of our two experiment datasets: “#S”, “#T”, “#A”, and “#O” denote the numbers of “Sentences”, “Triplets”, “Aspects”, and “Opinions”; “#S1”, “#S2”, “#S3” denote the numbers of sentiments “Positive”, “Neutral” and “Negative”, respectively.

proof in **Appendix Proof 2**.

Theorem 1. From insight of the above lemmas, it can be concluded that using **enough** (that is, following the 1-1 map properties in Lemma 1, as well as avoiding the issues in Lemma 2) labels will make it a theoretically ensured tagging scheme.

Assumption 1. Ceteris paribus, for a specific classification neural network, the **fewer** the number of target categories, the easier it is for the network to learn. This is an empirical and heuristic assumption, for the reasonable consideration of *Simplification of Decision Boundaries* (Hinton and Salakhutdinov, 2006) and *Enhancement of Training Efficiency* (less parameters).

Combining Theorem 1 and Assumption 1, **fewer** yet **enough** labels can be heuristically better solution with theoretical guarantee.

With the above knowledge, our tagging scheme employs a full matrix (illustrated as Figure 6) so that rectangular occupations in its cells indicate respective triplets, where each of the rectangles’ row indices correspond to the relative Aspect term and the column indices correspond to the Opinion. Hereafter, this kind of labels can be taken as a set

with the word span.

of “place holder”, which is obviously a 1-1 map meeting Lemma 1.

To further satisfy Lemma 2, we introduce another kind of labels, “sentiment & beginning tag”. This set of labels specializes in recognizing the top-left corner of a “shadowed” area. Meanwhile, it takes a value from the sentiment polarity, i.e. Positive, Neutral, Negative. This tagging is crucial to both *identify the beginning of an triplet* and *label the sentiment polarity*.

Figure 6 shows a comprehensive case of our tagging scheme, in which the left matrix is an appearance of our tagging scheme, and it can be decomposed into two separate components. The middle matrix is the first component, which takes only one tag to locate the up-left beginning of an area, and the second component simply predicts a binary classification to figure out the full area.

Note that, this design benefits the tagging scheme’s decode process. By scanning across the matrix, we only start an examination function when triggered by a beginning label like this, and then search by row and column until it meets any label except a “continued” (“CTD”), which satisfies Lemma 2.

A.4 Proof 1:

Let:

- S be a sentence with n tokens.
- M be an $n \times n$ tagging matrix for S , where each entry $M[i][j]$ can hold a label.
- $T_k = (A_k, O_k, S_k)$ be a sentiment triplet consisting of an aspect term A_k , an opinion term O_k , and a sentiment S_k .

Tagging Strategy If A_k starts at position i and O_k starts at position j , then $M[i][j]$ is tagged with a unique label L_k that encodes S_k . This label L_k uniquely identifies the triplet T_k , ensuring that no other entry $M[i'][j']$ with $(i', j') \neq (i, j)$ carries the same label unless it refers to the same sentiment context.

Define $L_k = \text{"start of triplet"}T_k$ with sentiment S_k

Proof of One-to-One Mapping

- **Injectivity:** Each L_k uniquely identifies a triplet T_k . If $M[i][j] = M[i'][j'] = L_k$, then by definition, $(i, j) = (i', j')$ and T_k is the same.

- **Surjectivity:** Each triplet T_k can be uniquely located and identified by its label L_k in matrix M , where no two distinct triplets have the same label at the same matrix position.

Conclusion The tagging scheme ensures that each sentiment triplet T_k is uniquely mapped to a specific label in the matrix M , and each label in M uniquely refers back to a specific triplet T_k . This guarantees a one-to-one correspondence between the triplets and their tagging matrix representations, fulfilling the conditions required by Lemma 1 for an effective and efficient ASTE process.

A.5 Proof 2:

For the ASTE task, considered as a 2D-labeling problem, it is necessary to ensure three fundamental goals are met:

Definitions

- S be a sentence with n tokens.
- M be an $n \times n$ tagging matrix for S , where each entry $M[i][j]$ can hold a label indicating a component of a sentiment triplet.
- $T_k = (A_k, O_k, S_k)$ be a sentiment triplet consisting of an aspect term A_k , an opinion term O_k , and a sentiment S_k .

Goals

1. **Correct Identification of Pairs:** Ensure that each (Aspect, Opinion) pair is correctly identified in the tagging matrix M .
2. **Classification of Sentiment Polarity:** Accurately classify the sentiment polarity S_k for each (Aspect, Opinion) pair.
3. **Avoidance of Boundary Errors:** Prevent boundary errors such as overlapping and confusion in the tagging matrix M .

Proof Using Contraposition

1. **Assuming Incorrect Identification:** Assume that some (Aspect, Opinion) pairs are incorrectly identified in M . This would mean that there exists at least one pair (i, j) where $M[i][j]$ does not represent the actual (Aspect, Opinion) relationship in S . This misrepresentation leads to incorrect sentiment analysis results, which contradicts the requirement of the task to provide accurate sentiment analysis, thereby proving that our identification must be correct.

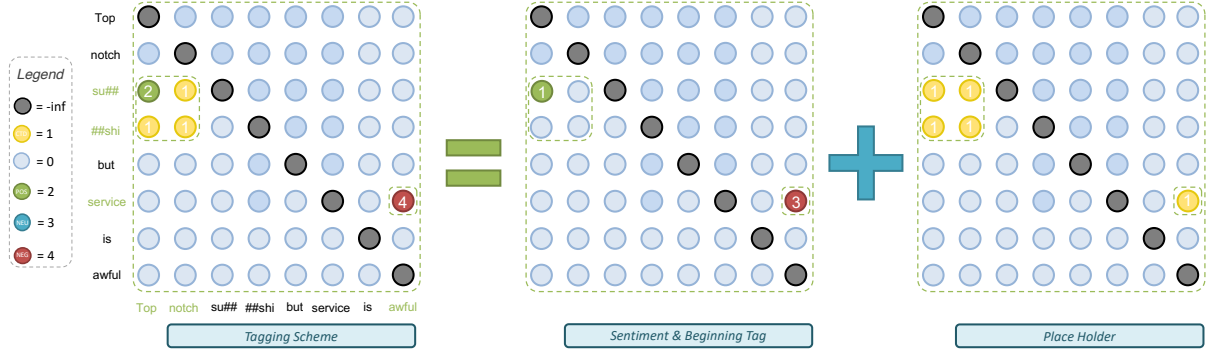


Figure 6: Decomposition of the tagging scheme into two components: 1) a beginning mark matrix with sentiment labels; and 2) a placeholder matrix denoting regions of triplets with “1”s and default regions with “0”s. Remember that each row is taken as candidates for an Aspect and each column is taken as candidates for an Opinion. Naturally, each cell in the square matrix can be seen as an ordered pair for a unique candidate of <Aspect, Opinion>. When we simply sum the two components up, we have the left-hand tagging scheme in Figure 6, where the “Sentiment & Beginning Tag” is like a trigger (just like you click your mouse), and the “Place Holder” is like a “continued shift” (continue to hold and drag the mouse to the downright).

2. **Assuming Incorrect Classification:** Assume the sentiment polarity S_k is incorrectly classified in M . This would imply that the sentiment associated with an (Aspect, Opinion) pair is wrong, leading to a sentiment analysis that does not reflect the true sentiment of the text. Given that the primary goal of ASTE is to accurately identify sentiments, this assumption leads to a contradiction, thereby establishing that our classification must be accurate.
3. **Assuming Existence of Boundary Errors:** Assume boundary errors such as overlaps or confusion occur in M . Such errors would prevent the clear identification and classification of sentiment triplets, leading to incorrect or ambiguous extraction outcomes. This would undermine the integrity and usability of the ASTE process, contradicting the task’s need for precise extraction mechanisms. Hence, we prove that boundary errors must be effectively managed.

A.6 Baselines

See Table 6.

A.7 Performance on Other ABSA Tasks

See Table 7.

A.8 ASTE Performance on ($\mathcal{D}_1$)

See Table 8.

A.9 Details Settings of GPT Experiments

See Table 9.

A.10 Detailed Results of GPT Experiments

See Table 11.

A.11 Case Study

See Table 12.

Conclusion The contraposition approach solidifies that the tagging strategy for ASTE in a 2D labeling framework successfully achieves the correct identification of pairs, accurate classification of sentiment, and effective management of boundary errors, as any failure in these aspects leads to contradictions with the task requirements.

Methods	Brief Introduction
Pipeline	
OTE-MTL (Zhang et al., 2020)	It proposes a multi-task learning framework including two parts: aspect and opinion tagging, along with word-level sentiment dependency parsing. This approach simultaneously extracts aspect and opinion terms while parsing sentiment dependencies using a biaffine scorer. Additionally, it employs triplet decoding based on the aforementioned outputs during inference to facilitate triplet extraction.
Li-unified-R+PD (Peng et al., 2020)	It proposes an unified tagging scheme, Li-unified-R, to assist target boundary detection. Two stacked LSTMs are employed to complete aspect-based sentiment prediction and the sequence labeling.
CMLA+C-GCN (Wang et al., 2017)	It facilitates triplet extraction by modelling the interaction between the aspects and opinions.
Two-satge (Peng et al., 2020)	It decomposes triplet extraction to two stages: 1) predicting unified aspect-sentiment and opinion tags; and 2) pairing the two results from stage one.
RI-NANTE+ (Dai and Song, 2019)	It adopts the same sentiment triplets extracting method as that of CMLA+, but it incorporates a novel LSTM-CRF mechanism and fusion rules to capture word dependencies within sentences.
Sequence-tagging	
Span-BART (Yan et al., 2021)	It redefines triplet extraction within an end-to-end framework by utilizing a sequence composed of pointer and sentiment class indexes. This is achieved by leveraging the pretrained sequence-to-sequence model BART to address ASTE.
JET (Xu et al., 2020)	It extracts triplets jointly by designing a position-aware sequence-tagging scheme to extract the triplets and capturing the rich interactions among the elements.
Seq2seq	
Dual-MRC (Mao et al., 2021)	It proposes a solution for ASTE by jointly training two BERT-MRC models with parameters sharing.
BMRC (Chen et al., 2021a)	It introduces a bidirectional MRC (BMRC) framework for ASTE, employing three query types: non-restrictive extraction queries, restrictive extraction queries, and sentiment classification queries. The framework synergistically leverages two directions, one for sequential recognition of aspect-opinion-sentiment and the other for sequential recognition of opinion-aspects-sentiment expressions.
Table-filling	
GTS (Wu et al., 2020a)	It proposes a novel 2D tagging scheme to address ASTE in an end-to-end fashion only with one unified grid tagging task. It also devises an effective inference strategy on GTS that utilizes mutual indication between different opinion factors to achieve more accurate extraction.
Double-encoder (Jing et al., 2021)	It proposes a dual-encoder model that capitalizes on encoder sharing while emphasizing differences to enhance effectiveness. One of the encoders, referred to as the pair encoder, specifically concentrates on candidate aspect-opinion pair classification, while the original encoder retains its focus on sequence labeling.
S ³ E ² (Chen et al., 2021b)	It represents the semantic and syntactic relationships between word pairs, employs GNNs for encoding, and applies a more efficient inference strategy.
EMC-GCN (Chen et al., 2022)	It employs a biaffine attention module to embed ten types of relations within sentences, transforming the sentence into a multi-channel graph while incorporating various enhanced linguistic features to enhance performance. Additionally, the method introduces an effective strategy for refining word-pair representations, aiding in the determination of whether word pairs are a match or not.
LLM-based	
zero-shot	Performing aspect-based sentiment analysis using an LLM. The specific method involves inputting a prompted sentence and directly outputting the corresponding [A, O, S] triplets. An example of the text given to the LLM, with the prompt added, is as follows: "Perform aspect-based sentiment analysis on the provided text and return triplets as [Aspect, Opinion, Sentiment]. You only need to provide the triplets, no additional explanations are required. The provided text: {sentence}"
few-shot	Building upon the zero-shot method, a small number of examples from the training set are added to the prompted sentence: "Perform aspect-based sentiment analysis on the provided text and return triplets as [Aspect, Opinion, Sentiment]. For example: input: {train sentence} output: {train triplets}, ... (some other examples). You only need to provide the triplets, no additional explanations are required. The provided text: {sentence}". We utilized 5-shot, 10-shot, and 20-shot methods, all randomly sampled from the training set. The results indicate that the 5-shot method performed the best, while the performances of the 10-shot and 20-shot methods showed a decline. The tables presents the output results for the 5-shot method.

Table 6: Baseline methods with brief introduction.

Methods	14Res			14Lap			15Res			16Res		
	AE	OE	AOPE	AE	OE	AOPE	AE	OE	AOPE	AE	OE	AOPE
CMLA	81.22	83.07	48.95	78.68	77.95	44.10	76.03	74.67	44.60	74.20	72.20	50.00
RINANTE	81.34	83.33	46.29	77.13	75.34	29.70	73.38	75.40	35.40	72.82	70.45	30.70
Li-unified	81.62	85.26	55.34	78.54	77.55	52.56	74.65	74.25	56.85	73.36	73.87	53.75
GTS	83.82	85.04	75.53	79.52	78.61	65.67	78.22	79.31	67.53	75.80	76.38	74.62
Dual-MRC	86.60	86.22	77.68	80.44	79.90	63.37	75.08	77.52	64.97	76.87	77.90	75.71
MiniConGTS (Ours)	86.55	87.04	79.60	82.62	83.41	73.23	86.53	83.05	73.87	85.48	87.06	76.29
$\Delta F1$	-0.05	0.82	1.92	2.18	3.51	7.56	8.31	3.74	6.34	8.61	9.16	0.58

Table 7: F1-score performance on other ABSA tasks: AE, OE, and AOPE. The test is implemented on $\mathcal{D}_1$. Results of other models are retrieved from (Fei et al., 2022).

Methods	14Res			14Lap			15Res			16Res		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Pipeline												
OTE-MTL (Zhang et al., 2020)	-	-	45.05	-	-	59.67	-	-	48.97	-	-	55.83
Li-unified-R+PD [‡] (Peng et al., 2020)	41.44	68.79	51.68	42.25	42.78	42.47	43.34	50.73	46.69	38.19	53.47	44.51
RI-NANTE+ (Dai and Song, 2019)	31.42	39.38	34.95	21.71	18.66	20.07	29.88	30.06	29.97	25.68	22.30	23.87
CMLA+C-GCN [‡] (Wang et al., 2017)	72.22	56.35	63.17	60.69	47.25	53.03	64.31	49.41	55.76	66.61	59.23	62.70
Two-satge [‡] (Peng et al., 2020)	58.89	60.41	59.64	48.62	45.52	47.02	51.7	46.04	48.71	59.25	58.09	59.67
Sequence-tagging												
Span-BART (Yan et al., 2021)	-	-	72.46	-	-	57.59	-	-	60.10	-	-	69.98
JET (Xu et al., 2020)	67.97	60.32	63.92	58.47	43.67	50.00	58.35	51.43	54.67	64.77	61.29	62.98
MRC based												
BMRC [†] (Chen et al., 2021a)	71.32	70.09	70.69	65.12	54.41	59.27	63.71	58.63	61.05	67.74	68.56	68.13
COM-MRC (Zhai et al., 2022)	<u>76.45</u>	69.67	72.89	64.73	56.09	60.09	68.50	59.74	63.65	<u>72.80</u>	70.85	71.79
Table-filling												
S ³ E ² (Chen et al., 2021b)	69.08	64.55	66.74	59.43	46.23	52.01	61.06	56.44	58.66	71.08	63.13	66.87
GTS (Wu et al., 2020a)	70.92	69.49	70.20	57.52	51.92	54.58	59.29	58.07	58.67	68.58	66.60	67.58
EMC-GCN (Chen et al., 2022)	71.85	72.12	71.78	61.46	<u>55.56</u>	58.32	59.89	61.05	60.38	65.08	71.66	68.18
BDTF (Zhang et al., 2022b)	76.71	<u>74.01</u>	<u>75.33</u>	68.30	55.10	<u>60.99</u>	66.95	65.05	65.97	73.43	<u>73.64</u>	73.51
DGCNAP (Li et al., 2023)	71.83	68.77	70.26	66.46	54.34	58.74	62.03	57.18	59.49	69.39	72.20	70.77
LLM-based												
GPT-3.5-turbo zero-shot	39.21	56.17	46.18	26.21	40.69	31.88	31.21	52.75	39.21	35.28	59.64	44.34
GPT-3.5-turbo few-shot	50.32	64.75	56.63	29.67	43.90	35.41	36.94	61.01	46.02	44.80	69.96	54.62
GPT-3.5-turbo CoT	40.78	57.93	47.86	28.37	43.25	34.27	35.17	57.11	43.53	40.32	65.79	50.00
GPT-3.5-turbo CoT+few-shot	44.97	57.81	50.59	28.31	43.04	34.15	35.71	58.49	44.35	43.72	66.45	52.74
Ours												
MiniConGTS	75.87	76.12	76.00	<u>67.45</u>	61.01	64.07	<u>66.84</u>	<u>64.08</u>	<u>65.43</u>	69.38	74.40	<u>71.80</u>

Table 8: Experimental results on $\mathcal{D}_1$ (Wu et al., 2020a). The best results are highlighted in bold, while the second best results are underscored. The results with [†] are retrieved from (Yu Bai Jian et al., 2021). The results with [‡] are retrieved from (Xu et al., 2020). The results with [‡] are retrieved from (Wu et al., 2020a). The results with [‡] are retrieved from (Peng et al., 2020).

Sentence	Ground Truth	Predictions		
		GPT-3.5 zero-shot	GPT-3.5 few-shot	GPT-4 zero-shot
It is a cozy place to go with a couple of friends.	[place, cozy, positive]	[place, cozy, positive], [friends, couple of, positive]	[place, cozy, positive]	[place, cozy, positive]
This place has the best sushi in the city.	[sushi, best, positive]	[sushi, best, positive], [place, best, positive], [city, best, positive]	[sushi, best, positive]	[sushi, best, positive], [place, This, neutral], [city, in the, neutral]
Disappointingly, their wonderful Sakerini has been taken off the bar menu.	[Sakerini, wonderful, positive], [bar menu, Disappointingly, negative]	[Sakerini, wonderful, positive], [bar menu, taken off, negative]	[Sakerini, taken off, negative]	[Sakerini, wonderful, positive], [bar menu, taken off, negative]
I went in one day asking for a table for a group and was greeted by a very rude hostess.	[hostess, rude, negative]	[hostess, rude, negative]	[hostess, rude, negative]	[table for a group, asking, neutral], [hostess, very rude, negative]
But make sure you have enough room on your credit card as the bill will leave a big dent in your wallet.	[bill, big, negative]	[credit card, enough room, positive], [bill, big dent, negative], [wallet, big dent, negative]	[credit card, enough room, negative], [bill, big dent, negative], [wallet, big dent, negative]	[bill, big dent, negative]

Table 9: In summary, there are several challenges observed in the performance of GPT models concerning triplets. Firstly, there is a prominent issue of "hard" matching, where GPT models tend to introduce additional modifiers or adverbs in the opinion component, leading to a lack of exact correspondence. Secondly, during zero-shot inference, GPT models tend to generate multiple predicted triplets, resulting in decreased precision. This behavior particularly hampers the precision of the model's predictions. Thirdly, inconsistencies arise in handling triplets involving structures such as [A, O1 and O2, S] and [A, O1, S], [A, O2, S]. This inconsistency is challenging to mitigate due to its dependence on annotation practices and conventions. Upon closer examination, the issues observed do not appear to be as pronounced as indicated by the evaluation metrics. Rather, they often manifest as cases where the general idea is correctly captured, but the precise format or phrasing does not align perfectly. Notably, the performance of GPT-4 deteriorates due to its occasional tendency to not merely "extract" fragments from sentences but to generate its own summarizations. Consequently, evaluating against triplets that originate solely from annotated sentences poses a challenge in achieving alignment. Furthermore, GPT-4 exhibits a proclivity for extracting longer sequences of words as aspects or opinions, while GPT-3.5 tends to produce shorter sequences that better conform to typical annotation scenarios.

Methods	Prompts
zero-shot	Suppose you are an expert of aspect-based sentiment analysis. Perform aspect-based sentiment analysis on the provided text and return triplets as [Aspect, Opinion, Sentiment]. You only need to provide the triplets, no additional explanations are required. The provided text: {sentence}
few-shot	Suppose you are an expert of aspect-based sentiment analysis. Perform aspect-based sentiment analysis on the provided text and return triplets as [Aspect, Opinion, Sentiment]. For example: input: The food is uniformly exceptional , with a very capable kitchen which will proudly whip up whatever you feel like eating , whether it 's on the menu or not . output: ['food', 'exceptional', 'positive'], ['kitchen', 'capable', 'positive'] ... (generated from training set) Now I will provide a new sentence, and you only need to provide the triplets [Aspect, Opinion, Sentiment] without any additional explanations. The provided sentence: {sentence}
CoT	Suppose you are an expert of aspect-based sentiment analysis. Please analyze the given text for aspect-based sentiment analysis using the following steps: Definitions: - Aspect: An aspect is a specific part or feature of the entity being discussed. It is usually a noun or a noun phrase. - Opinion: An opinion is a descriptive term or phrase that expresses a sentiment towards the aspect. It is usually an adjective or a descriptive phrase. - Sentiment: The sentiment is the overall feeling expressed towards the aspect, categorized as positive, negative, or neutral. Instructions: 1. Read the text and identify all aspects mentioned. 2. For each identified aspect, determine the opinion expressed and the sentiment (positive, negative, neutral). 3. Summarize the findings in the format [Aspect, Opinion, Sentiment]. Each triplet must contain an aspect, an opinion, and a sentiment. 4. If there is a one-to-many relationship between aspects and opinions, list multiple triplets. 5. Use all words from the original text to answer without any changes. Example: (automatically generated by ChatGPT-4o) Text: "The restaurant has a great ambiance, but the service is poor and the food is average." Steps: 1. Identify aspects: ambiance, service, food. 2. Evaluate opinions and sentiments: - ambiance: Opinion - great, Sentiment - positive - service: Opinion - poor, Sentiment - negative - food: Opinion - average, Sentiment - neutral 3. Summarize: - [ambiance, great, positive] - [service, poor, negative] - [food, average, neutral] Please analyze the following text: {sentence}
CoT+few-shot	Suppose you are an expert of aspect-based sentiment analysis. Please analyze the given text for aspect-based sentiment analysis using the following steps: Definitions: - Aspect: An aspect is a specific part or feature of the entity being discussed. It is usually a noun or a noun phrase. - Opinion: An opinion is a descriptive term or phrase that expresses a sentiment towards the aspect. It is usually an adjective or a descriptive phrase. - Sentiment: The sentiment is the overall feeling expressed towards the aspect, categorized as positive, negative, or neutral. Instructions: 1. Read the text and identify all aspects mentioned. 2. For each identified aspect, determine the opinion expressed and the sentiment (positive, negative, neutral). 3. Summarize the findings in the format [Aspect, Opinion, Sentiment]. Each triplet must contain an aspect, an opinion, and a sentiment. 4. If there is a one-to-many relationship between aspects and opinions, list multiple triplets. 5. Use all words from the original text to answer without any changes. Example: (generated from training set) Text: ... Steps: 1. Identify aspects: ... 2. Evaluate opinions and sentiments: - ... 3. Summarize: - [..., ..., ...] ... Please analyze the following text: {sentence}

Table 10: LLM prompts in different methods.

Method	Combination	14Res			14Lap			15Res			16Res		
		P	R	F1	P	R	F1	P	R	F1	P	R	F1
GPT-3.5-turbo few-shot	A-O-S	0.5151	0.6519	0.5755	0.3979	0.5009	0.4435	0.4334	0.6309	0.5139	0.5112	0.7101	0.5945
	A-O	0.5429	0.6871	0.6066	0.4479	0.5638	0.4992	0.4788	0.6969	0.5676	0.5420	0.7529	0.6303
	A-S	0.6234	0.7983	0.7001	0.4955	0.7149	0.5853	0.5539	0.7731	0.6454	0.5853	0.8274	0.6856
	O-S	0.5790	0.7438	0.6512	0.5016	0.6653	0.5719	0.5228	0.7281	0.6086	0.6020	0.7851	0.6814
	A	0.6758	0.8455	0.7512	0.5805	0.8337	0.6844	0.6220	0.838	0.7140	0.6361	0.8739	0.7363
	O	0.6138	0.7938	0.6923	0.5587	0.7505	0.6406	0.5780	0.8048	0.6728	0.6444	0.8404	0.7295
	S	0.8251	0.9222	0.8710	0.7907	0.8844	0.8349	0.8286	0.9337	0.878	0.8179	0.9466	0.8776
GPT-3.5-turbo CoT	A-O-S	0.4847	0.5905	0.5324	0.3048	0.4030	0.3471	0.3951	0.5670	0.4657	0.4403	0.6381	0.5210
	A-O	0.5178	0.6308	0.5687	0.3566	0.4713	0.4061	0.4368	0.6268	0.5148	0.4711	0.6829	0.5576
	A-S	0.5991	0.7807	0.6779	0.4088	0.6199	0.4927	0.5123	0.7685	0.6148	0.5199	0.8075	0.6326
	O-S	0.5708	0.7143	0.6345	0.4288	0.5768	0.4919	0.4834	0.6711	0.5620	0.5481	0.7511	0.6338
	A	0.6594	0.8538	0.7441	0.5142	0.7797	0.6197	0.5901	0.8796	0.7063	0.5702	0.8805	0.6922
	O	0.6119	0.7723	0.6828	0.4930	0.6716	0.5686	0.5403	0.7500	0.6281	0.5932	0.8128	0.6858
	S	0.7942	0.9374	0.8599	0.7229	0.9046	0.8036	0.7477	0.9222	0.8258	0.7494	0.9585	0.8411
GPT-3.5-turbo CoT+few-shot	A-O-S	0.4941	0.5915	0.5385	0.3378	0.4233	0.3757	0.3902	0.5608	0.4602	0.4649	0.6693	0.5486
	A-O	0.5294	0.6338	0.5769	0.3968	0.4972	0.4413	0.4333	0.6227	0.5110	0.4932	0.7101	0.5821
	A-S	0.6257	0.7925	0.6993	0.4518	0.6479	0.5324	0.5306	0.7824	0.6324	0.5451	0.8296	0.6579
	O-S	0.5735	0.7048	0.6324	0.4681	0.6168	0.5322	0.4936	0.6776	0.5712	0.5769	0.7745	0.6612
	A	0.6907	0.8691	0.7697	0.5572	0.7991	0.6566	0.6026	0.8704	0.7121	0.5873	0.8783	0.7039
	O	0.6186	0.7676	0.6851	0.5319	0.7100	0.6082	0.5479	0.7522	0.6340	0.6149	0.8255	0.7048
	S	0.8119	0.9336	0.8685	0.7470	0.8960	0.8147	0.7775	0.9366	0.8497	0.7923	0.9733	0.8735
GPT-4o few-shot	A-O-S	0.5411	0.6620	0.5955	0.3823	0.4861	0.4280	0.4557	0.6041	0.5195	0.5290	0.7101	0.6063
	A-O	0.5757	0.7042	0.6335	0.439	0.5582	0.4915	0.5023	0.6660	0.5727	0.5551	0.7451	0.6362
	A-S	0.6777	0.8208	0.7424	0.4962	0.7084	0.5836	0.5872	0.7639	0.6640	0.6019	0.8296	0.6977
	O-S	0.6047	0.7532	0.6709	0.4863	0.6337	0.5503	0.5613	0.7325	0.6356	0.6318	0.8106	0.7102
	A	0.7367	0.8679	0.7970	0.5951	0.8445	0.6982	0.6611	0.8218	0.7327	0.6471	0.8761	0.7444
	O	0.6389	0.8033	0.7117	0.5517	0.7271	0.6274	0.6185	0.8070	0.7003	0.6667	0.8553	0.7493
	S	0.8353	0.9336	0.8817	0.7758	0.8699	0.8202	0.8376	0.9366	0.8844	0.8346	0.9585	0.8923
GPT-4o CoT	A-O-S	0.4121	0.5332	0.4649	0.2698	0.3771	0.3146	0.3307	0.5093	0.4010	0.3914	0.5817	0.4679
	A-O	0.4331	0.5604	0.4886	0.3122	0.4362	0.3639	0.3614	0.5567	0.4383	0.4162	0.6187	0.4977
	A-S	0.6163	0.8278	0.7066	0.4486	0.6976	0.5461	0.5200	0.8125	0.6341	0.5374	0.8429	0.6563
	O-S	0.4711	0.6246	0.5371	0.3439	0.4779	0.4000	0.4093	0.5987	0.4862	0.4752	0.6532	0.5502
	A	0.6667	0.8703	0.7550	0.5280	0.8143	0.6406	0.5762	0.8750	0.6949	0.5799	0.8916	0.7027
	O	0.5004	0.6687	0.5724	0.3951	0.5544	0.4614	0.4558	0.6667	0.5414	0.5077	0.6979	0.5878
	S	0.7859	0.9545	0.8620	0.7651	0.9133	0.8327	0.7744	0.9597	0.8571	0.7725	0.9674	0.8590
GPT-4o CoT+few-shot	A-O-S	0.4681	0.5986	0.5254	0.2971	0.4085	0.3440	0.3508	0.5381	0.4247	0.4153	0.6109	0.4945
	A-O	0.4965	0.6348	0.5572	0.3454	0.4750	0.4000	0.3911	0.6000	0.4736	0.4437	0.6518	0.5280
	A-S	0.6295	0.8314	0.7165	0.4718	0.7235	0.5712	0.5262	0.8125	0.6388	0.5458	0.8429	0.6626
	O-S	0.5255	0.6824	0.5937	0.3890	0.5347	0.4504	0.4527	0.6513	0.5342	0.5102	0.6894	0.5864
	A	0.6795	0.8774	0.7658	0.5467	0.8337	0.6604	0.5898	0.8819	0.7069	0.5858	0.8916	0.7070
	O	0.5610	0.7342	0.6360	0.4395	0.6119	0.5116	0.5023	0.7215	0.5923	0.5465	0.7383	0.6281
	S	0.7927	0.9431	0.8614	0.7705	0.9220	0.8395	0.7890	0.9481	0.8613	0.7976	0.9703	0.8755

Table 11: Performance of different types of element combinations in ABSA tasks using LLM.

Sentence	Ground Truth	GPT results	Ours	Precision	Recall
Creamy appetizers-taramasalata, eggplant salad, and Greek yogurt(with cucumber, dill, and garlic) taste excellent when on warm pitas.	['creamy appetizers', 'creamy', 'positive'], ['creamy appetizers', 'excellent', 'positive'], ['warm pitas', 'warm', 'neutral'], ['taramasalata', 'creamy', 'positive'], ['eggplant salad', 'excellent', 'positive'], ['greek yogurt (with cucumber, dill, and garlic)', 'greek yogurt (with cucumber, dill, and garlic)', 'excellent', 'positive']	['appetizers', 'creamy', 'neutral'], ['taramasalata', 'taste excellent', 'positive'], ['eggplant salad', 'taste excellent', 'positive'], ['greek yogurt', 'taste excellent', 'positive'], ['warm pitas', 'warm', 'neutral']	['creamy appetizers', 'creamy', 'positive'], ['creamy appetizers', 'excellent', 'positive'], ['warm pitas', 'warm', 'neutral'], ['eggplant salad', 'excellent', 'positive'], ['greek yogurt', 'excellent', 'positive']	GPT: 1/5 Ours: 4/5	GPT: 1/6 Ours: 4/6
We left without ever getting service.	['service', 'without ever', 'negative']	['service', 'without ever getting', 'negative']	['service', 'without ever', 'negative']	GPT: 0/1 Ours: 1/1	GPT: 0/1 Ours: 1/1
I go out to eat and like my courses, servers are patient and never rush courses or force another drink.	['servers', 'patient', 'positive']	['courses', 'like', 'positive'], ['servers', 'patient', 'positive'], ['servers', 'never rush courses', 'positive'], ['servers', 'never force another drink', 'positive']	['servers', 'patient', 'positive'],	GPT: 1/5 Ours: 1/1	GPT: 1/1 Ours: 1/1

Table 12: Case study